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Studierichting: Informatica
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$$\int \frac{x^4+2x^3-12x^2+9}{x^3-3x+2}dx = \int \left(4x+2+\frac{-2x+5}{x^3-3x+2}\right)dx$$
$$= 2x^2+2x+\int \frac{-2x+5}{(x+2)\,(x-1)^2}dx$$
$$\rightarrow \frac{-2x+5}{(x+2)\,(x-1)^2} = \frac{A}{x+2}+\frac{B}{(x-1)^2}+\frac{C}{x-1}$$
$$\Rightarrow A = \frac{4+5}{(-2-1)^2} = \frac{9}{9} = 1$$
$$\Rightarrow B = \frac{-2+5}{3} = 1$$
$$\rightarrow \frac{-2x+5}{(x+2)\,(x-1)^2} - \frac{1}{(x-1)^2} = \frac{-3}{(x+2)\,(x-1)}$$
$$\Rightarrow C = \frac{-3}{3} = -1$$
$$= 2x^2+2x+\int \frac{1}{x+2}dx + \int \frac{1}{(x-1)^2}dx - \int \frac{1}{x-1}dx$$
$$= 2x^2+2x+\ln|x+2| - \frac{1}{x-1} - \ln|x-1| + c$$

References